

CAD FIT CHECK - will the real parts fit the printed parts?

[!] SUPERSEDED IN PART - read CAD_CHANGELOG_v7.5.md alongside this This document audited **v7.2/v7.3**. On 2026-07-30 a repair pass (**v7.5**) fixed every defect that did not require calipers. The findings below are preserved as the *record of what was wrong*, but the following are **FIXED and no longer describe the source**: | Section | Defect | Status |
|---|---|---| | §2a | Hall pocket entirely inside the cam pocket | [OK] rebuilt on the plate underside at (0, 17.35) | | §2c | Homing magnet craters the cam tracks | [OK] blind 1.2mm pocket, 3x1mm magnet (it craters **three** tracks, not two - §2c undercounts) | | §3a | Right motor screw opens into the spring cavity | [OK] spring cavity deleted; pilots instead of clearance holes | | §3b | Left ear hole 0.35mm from the plate edge | [OK] base_length 56 -> 58, wall now 1.35mm | | §5 | Muscle-board bosses clash with the ULN2003 | [OK] bosses now default OFF | | §6a | ESP32 overruns the pod by 1.75mm | [OK] pod_length 64 -> 68 | | §6c | USB cutout ~2mm too high | [OK] arithmetic bug fixed (ignored hdr_channel_depth) | | §6d | USB cutout too small for a real plug | [OK] 10x7 -> 13x9 | | §7a | Jack cradle overruns the cavity | [OK] fixed, plus an assert so it can't recur silently | **Still open, blocked on measurement** (see MEASUREMENTS_NEEDED.md): §4c cam z-stack - §2b hall sensor thickness - §7b real jack dimensions - §6b **M21 header-row pitch - 30-pin clones still vary, and this decides whether the ESP32 fits.**

2026-07-31 measurement-research update The online and owned-part pass is recorded in MEASUREMENT_RESEARCH.md. - Motor M1, M2, M3, M4, M6, M8, and M9 are now physically measured; M7b is derived. - The 49E/SS49E package envelope is published, but its 1.68 mm worst-case thickness exceeds the current 1.6 mm pocket, so M11b and the package marking remain assembly checks. - M18 is resolved as a 30-pin USB-C ESP32. M21 remains open because 30-pin boards are sold with 22.86, 25.4, and 27.94 mm header-row spacing. - The owned yellow/black barrel adapter is inline, not panel-mount. Final CAD should use a specified panel-mount 5.5 x 2.1 mm jack or be redesigned to capture the entire inline body.

Date: 2026-07-29 **Scope:** physical fit of purchased/owned components inside the v7.2 printed parts. **Status of the build:** nothing has ever been assembled. This is the last check before spending.

How to read this document

Every statement is tagged:

- **[SCAD]** - read directly out of the OpenSCAD source in cad/scad/. This is fact about the *printed part*.
- **[SPEC]** - a published nominal figure for a standard catalogue part (28BYJ-48, 6x6 tactile switch, 5.5/2.1 barrel jack, micro-USB). Standard parts still vary between clone vendors.
- **[ASSUMED]** - a number somebody typed into the CAD about a physical object that has never been measured. Treat as fiction until a caliper says otherwise.

Historical note: when this v7.2/v7.3 audit was written, the "Real part needs" column was unmeasured. Several owned-part dimensions have since been measured; use MEASUREMENT_RESEARCH.md as the current evidence table.

1. SUMMARY TABLE

Component	CAD provides [SCAD]	Real part needs	Verdict	Action
Hall sensor (MH-Sensor-Series / KY-024 module)	5.3 x 4.3 x 3.0 mm pocket at (0, +20) in base_plate	PCB module ~15 x 11 x 1.6 mm + ~11 mm of 4-pin header [ASSUMED]	FAILS	Desolder the bare sensor, or move the module out of the plate entirely
Hall pocket itself	pocket floor z=2; dia 46 cam pocket floor also z=2, and the pocket lies entirely inside r=23	any pocket at all	FAILS	The pocket is 100% subsumed by the cam pocket - it does not exist as a feature
Cam homing magnet dia 3x2	dia 3.2 x 2.0 pocket at r=17.35 in a 2.0 mm disc floor	3x2 mm disc magnet - not owned	FAILS	Pocket is a through-hole that punches a dia 3.2 crater through cam tracks 2 and 3
28BYJ-48 motor body	dia 29 seat, 35 mm hole spacing, dia 4.3 holes, body centre at x=-8, 19 mm of z-space	body dia 28, 35 mm spacing, dia 4.2, ~8 mm offset, 19 mm can [SPEC]	UNVERIFIED	Never measured. 8 caliper readings required (§4)

Component	CAD provides [SCAD]	Real part needs	Verdict	Action
Motor right mounting screw	hole at (+9.5, 0) falls inside the 22 x 16 spring-cavity through-window	material to clamp against	FAILS	Motor can only be held by ONE screw
Motor screw head vs cam	left hole at x=-25.5 is clear; right hole at x=+9.5 is under the dia 46 cam disc	screw head clearance	FAILS	No counterbore modelled; head fouls the cam floor
Cam on shaft (double-D bore)	dia 5.2 clipped to 3.2 flats; through-hole , real engagement = hub 4 + floor 2 = 6 mm	5.0 mm shaft, 3.0 mm across flats, ~10 mm long [SPEC]	UNVERIFIED / marginal	Clearances fine on paper; z-stack is wrong (\$5)
Cam z-stack vs motor face	hub bottom z=39, motor top face z=41	hub cannot go below the motor face	FAILS	Cam bottoms on the motor and sits ~2 mm high -> all six dots permanently proud
Shaft protrusion	cam surface z=45; a 10 mm shaft reaches z=51	shaft must not enter the dot cluster	FAILS (conditional)	Shaft passes through the open bore into the linkage risers at x=+/-2.4
ULN2003 module, wires soldered flat	36 x 46 x 16 pocket + 2 mm mid-plate relief slot = 18 mm	~35 x 32 x ~12 mm [ASSUMED, established earlier]	FITS	Solder wires flat. Watch the 0.5 mm/side X clearance
ULN2003 module, vertical Dupont jumpers	same 18 mm max	~20 mm as-wired [ASSUMED]	FAILS	Confirmed: 20 > 18. No vertical jumpers
ULN2003 vs muscle-board bosses	4 mm tall bosses at (+/-14, +/-19) rise from the pocket floor	flat floor	UNVERIFIED	If the board is ~35 mm in Y it clips the y=+/-19 bosses
ESP32 DevKit length in pod	cavity +/-28 in X; board 51.5 centred at x=+4 -> spans -21.75 .. +29.75	board must end inside x=+28	FAILS	1.75 mm interference with the +X (dock) wall
ESP32 DevKit width / row pitch	devkit_width 28.0, hdr_row_pitch 25.4 [ASSUMED]	30-pin DOIT V1 is commonly ~25.4 wide / 0.9" pitch; 38-pin is ~28 / 1.0" [SPEC-ish]	UNVERIFIED	The CAD numbers describe the 38-pin board; the docs say 30-pin
USB cutout alignment	cutout z 12.5 .. 19.5; board top sits at z=10.5 (8.5 strip ? 1.0 channel recess)	port body ~10.5 .. 13.0	FAILS	Hole sits ~2 mm too high; only ~0.5 mm overlaps the port
USB cutout size	10 x 7 mm	micro-USB overmould commonly ~12 x 8 [SPEC-ish]	UNVERIFIED	Plug may not enter
Headroom under pod lid	board top z~10.5, cavity top z=54	a few mm	FITS	43 mm spare - grossly over-tall, but no clash
Barrel jack hole	dia 11.5 at (-22, +18) on the lid	5.5/2.1 panel jack, ~dia 11 thread [SPEC]	UNVERIFIED	User's jack is a yellow screw-terminal type, not the modelled panel jack
Barrel jack cradle	inner 12 x 9.5 x 11, walls 1.5 -> outer reaches x = -29.5	must fit inside cavity x >= -28	FAILS	1.5 mm interference - the lid will not close. And all three dims are PLACEHOLDERS
Docking magnets	dia 8.4 x 1.2 pockets, 2 per face at y=+/-14, 4 mm wall	8 x 1 mm NdFeB discs - measured, in hand	FITS	Only re-confirm the newly bought batch is the same 8x1
Nav buttons - cap stem	dia 3.8 stem projects 4.5 mm behind the flange through a dia 4.2 hole	must cross a 4 mm wall	FITS (nominal)	Stem reaches 0.5 mm inside; 0.4 mm diametral clearance allows resin-in-PETG sliding
Nav buttons - switch reach	The 6 mm switch body is stopped by the pod's inner front wall; cage back wall takes press load	plunger must meet the stem	FITS (nominal)	The front wall is the forward datum; bench-test the actual switch's plunger before final resin batch

Component	CAD provides [SCAD]	Real part needs	Verdict	Action
6x6x5 tactile switch in cage	6.4 mm slot, 7.2 mm depth, legs vertical	6.0 x 6.0 x 5.0 body [SPEC]	FITS	Body and revised cap reach fit on the nominal switch geometry
Dot insert into top plate	body 11.0 into 11.2 opening; flange 15.0 into 15.2 rebate; heights 2.0 / 1.2; glue shelf 2.0 mm wide, 105.6 mm2	resin tile printed from the same file	FITS	Verified numerically. Only risk is FDM square holes printing undersize
Base-plate standoffs	dia 6 posts at x=+-26, plate half-length 28 -> 1 mm hangs off the edge	supported footprint	UNVERIFIED / cosmetic	Prints as a small unsupported overhang; not structural

Required outer-box changes - handout checklist

These items are deliberately **not** applied to outer_box.scad yet. They either require the real motor dimensions or a hardware-choice decision; changing them from catalogue numbers would simply move the fit failure somewhere else.

Priority	Required change	Why	Prerequisite
Blocker	Re-derive base_plate_z and the motor-height allowance	The current 41 mm motor-face height is an unmeasured assumption. It controls the cam/linkage vertical stack.	Measure the actual motor can height and shaft height.
Blocker	Redesign the base-plate spring cavity around both motor ears	The right motor screw opens into the existing 22 x 16 mm through-window, leaving a one-screw mount.	Measure the actual motor mounting-ear envelope and hole centres.
Blocker	Widen/reposition the base-plate support footprint if the motor-ear fix needs it	The left ear currently has only a 0.35 mm edge wall and the base standoffs overhang the plate by 1 mm.	Motor-ear measurement, then confirm the revised plate still clears the 60 mm shell cavity and corner bosses.
Required before installing the owned Hall module	Add a real Hall-module bracket in the outer-box electronics area, or commit to a measured bare-sensor design	The current base-plate pocket is entirely subsumed by the cam pocket and cannot hold the owned module.	Decide module versus desoldered sensor; measure its footprint and sensing-face offset.
Decision required	Remove/relocate the four future-muscle-board bosses when using the ULN2003 module	A 35 mm-wide ULN2003 can collide with the y=+-19 mm bosses in the outer-box electronics pocket.	Measure the actual ULN2003 board and choose prototype driver versus future muscle PCB.

Do not alter the 68 x 68 mm shell, docking windows, magnet pockets, or 4 mm walls as part of this repair.

Those features are independent of the motor and have already been physically fit-tested. The next safe sequence is: measure hardware -> update the dependent base/mid/cam/ outer-box dimensions together -> render the full assembly -> print a cheap proving part.

2. HALL SENSOR - FAILS, twice over

2a. The pocket does not physically exist

[SCAD] base_plate.scad:

- hall_pocket_h = 3, cut from base_thickness - hall_pocket_h = **z = 2** upward.
- cam_pocket_depth = 3, cam_pocket_diameter = 46, cut from **z = 2** upward, radius 23.
- Hall pocket footprint: x ? [-2.65, +2.65], y ? [17.85, 22.15]. Its farthest corner is at

radius ?(2.652 + 22.152) = **22.31 mm < 23 mm**.

The hall pocket lies **entirely inside the cam pocket, at exactly the same depth**. Subtracting it removes nothing. There is no rectangular recess on the printed part - just flat cam-pocket floor.

Worse: the cam disc's underside *sits on that floor* (cam_flat_z = 45, disc floor 2 mm, pocket floor z=43 in box coordinates). There is **zero vertical space** above the "pocket" for any sensor, bare TO-92 or otherwise. The wire channel (hall_wire_w=2 x hall_wire_d=1) does survive, but only in the strip of plate outside r=23.

2b. The module is far too big anyway

[ASSUMED - MEASURE THIS] The blue MH-Sensor-Series / KY-024-style board Mridul owns (identified by Codex on 2026-06-03 as an analog+digital module with AO/DO/GND/VCC) is a PCB in the region of **15 x 11 mm, ~1.6 mm thick, plus a 4-pin 2.54 mm header adding roughly 8-11 mm of height**. Those figures are typical for that family of modules; they are **not** measured from his part.

Against a 5.3 x 4.3 x 3.0 pocket sized for a bare SS49E/A3144 TO-92 body:

- length: needs ~15, has 5.3 -> **~10 mm short**
- width: needs ~11, has 4.3 -> **~7 mm short**
- height: needs ~10-13 with the header, has 3 (and really has 0, see 2a) -> **hopeless**

Fixes, in order of preference

1 Desolder the bare sensor from the module (recommended). The MH-series board carries a standard TO-92 hall device plus an LM393 comparator, an LED and a trimpot. Unsolder the three legs, mount the bare TO-92 in the plate, and run three wires back to the module (or straight to the ESP32 with a pull-up, since the breadboard test already ran the analog output at 3.3 V). This keeps the CAD footprint honest - but §2a still has to be fixed first, because the pocket currently isn't there.

1 Move the module off the base plate entirely. Mount it on the electronics-pocket floor or on a wall of the outer box, with the sensing face pointing up at the cam, and run wires. The cam magnet is 3 mm dia - flux falls off fast, so this only works if the module can get within a few mm of the magnet path. This is the lowest-CAD-risk option but needs a new bracket.

1 Enlarge and deepen the pocket. Requires all three of: (a) make `hall_pocket_h` > 3 so the pocket floor drops below the cam pocket floor (e.g. 4.0-4.5 mm, leaving 0.5-1.0 mm of plate underneath); (b) grow `hall_pocket_w/d` to the measured module footprint + 0.4 mm; (c) move `hall_pocket_y` outboard past `y=23` so it is no longer under the disc, **or** add a header escape slot. Note the plate is only 5 mm thick and the +Y edge is at `y=25` - there is very little room. Option 1 or 2 is more realistic.

2c. BONUS DEFECT FOUND - the cam homing-magnet pocket is a hole through two tracks

[SCAD] `braille_cam.scad`: `magnet_dia=3.0`, `magnet_depth=2.0`, cut as `cylinder(d=3.2, h=3)` from `z=-1`, i.e. `z = -1 .. +2`, at `r = 17.35`, angle 90deg. The disc floor is `disk_base_thickness = 2.0`, spanning `z = 0 .. 2`. The cut therefore removes the **entire floor thickness** - it is a through-hole, not a pocket ("leaves 0mm floor" is even in the comment). And at `r = 17.35 +/- 1.6` it spans `r = 15.75 .. 18.95`, which overlaps **track 2** (`r 15.4-17.0`, drives dot 1) and **track 3** (`r 17.1-18.7`, drives dot 2).

Where those tracks are in their DOWN state the cam surface at that angle is simply **missing** - a dia 3.2 mm hole in a running surface that a 1.0 mm-thick linkage foot crosses once per revolution. The foot will drop in and jam.

Fix: shrink `magnet_depth` to ~1.4 mm and thicken the local floor, or (better) move `magnet_radius` inboard to `r < 12` (inside `inner_radius`, under no track at all) and move the hall sensor to match. Moving the magnet inboard also weakens coupling, so re-check the sensor gap. Also: **the 3x2 mm homing magnet is not in hand** (`docs/SOURCING.md §2`) - it still has to be bought.

3. MOTOR - NEVER MEASURED. Treat every motor number as fiction.

[!] **THE 28BYJ-48 IN THIS PROJECT HAS NEVER BEEN TOUCHED WITH A CALIPER.** `base_plate.scad`, `mid_plate.scad` and `braille_cam.scad` all encode motor dimensions that were typed from datasheets and forum posts. Mridul has photographed a motor whose shaft looked different from the model. Every verdict in this section is conditional on measurement.

[SCAD] what the CAD assumes:

Parameter	Value	File
body diameter	dia 29 (seat pocket, 1.5 mm deep)	<code>base_plate.scad</code>
retaining collar ID	dia 29.5, 8 mm tall, at <code>x=-8</code>	<code>mid_plate.scad</code>
mount-hole spacing	35 mm	<code>base_plate.scad</code>
mount-hole diameter	dia 4.3	<code>base_plate.scad</code>
shaft offset from body centre	-8 mm (shaft at <code>x=0</code> , body at <code>x=-8</code>)	<code>base_plate.scad</code> , <code>mid_plate.scad</code>

Parameter	Value	File
can height allowance	19 mm (z=22 .. 41)	outer_box.scad base_plate_z
shaft	dia 5.0, 3.0 across flats, 10 mm long	braille_cam.scad comments

[SPEC] The published nominal 28BYJ-48 is: can dia 28 x 19 mm, mounting bracket holes 35 mm apart at dia 4.2, shaft offset ~8 mm from the can centre, shaft dia 5.0 with two flats ~3.0 mm apart, protruding ~10 mm. So the CAD is *plausible*. Clone variance and the 19 mm z-budget with **zero** slack are the risks.

Two hard failures that exist regardless of measurement

3a. The right-hand motor screw has nothing to screw into. [SCAD] `spring_cavity()` is `cube([22, 16, 20], center=true)` - a through-window spanning x ? [-11, +11], y ? [-8, +8]. The right mounting hole is at x = -8 + 35/2 = **+9.5**, y = 0, dia 4.3 -> spans x ? [7.35, 11.65]. That is inside the window. There is no plate material there. **The motor can only be bolted by its left ear (x=-25.5).** A single-screw stepper will rock and lose steps.

3b. The left ear hole is 0.35 mm from the plate edge. x = -25.5, dia 4.3 -> outer edge at -27.65; plate half-length is 28. A 0.35 mm wall on a 0.4 mm nozzle is not a wall. The screw will break out.

Fix for both: shrink/reshape `spring_cavity` so it clears x=+9.5 (e.g. two smaller windows, or move it to -Y), and widen `base_length` from 56 to at least 60 (the box cavity is 60 - so this also needs the cavity or the standoff pattern revisited).

Measurements required (feeds -> CAD parameter)

- 1 Can outer diameter -> `motor_body_diameter` (`base_plate`), collar ID (`mid_plate`)
- 2 Can height, bracket face to bottom -> `base_plate_z` in `outer_box.scad`
- 3 Shaft centre to can centre -> `motor_body_x_offset`, `motor_x_offset`
- 4 Shaft diameter -> cam bore dia (currently 5.2)
- 5 Shaft flat-to-flat -> cam bore flat clip (currently 3.2)
- 6 Shaft length above the bracket face -> `cam_hub_h` + `shaft_bore_depth` + z-stack
- 7 Mount-hole centre-to-centre -> `motor_mount_spacing`
- 8 Mount-hole diameter -> `motor_mount_hole`
- 9 Shaft boss (raised collar around the shaft) diameter and height -> cam hub OD (currently dia 9)

4. CAM ON SHAFT - clearances OK on paper, z-stack is wrong

4a. The bore profile

[SCAD] `braille_cam.scad`: `intersection(cylinder(r=2.6), cube([10, 3.2, ...]))` -> a dia 5.2 cylinder clipped to 3.2 mm across the flats, on both sides (true double-D).

[SPEC] Nominal 28BYJ-48 shaft: **dia 5.0 mm, 3.0 mm across the flats, ~10 mm long.**

- radial clearance: $(5.2 - 5.0)/2 = 0.10$ mm per side
- flat clearance: $(3.2 - 3.0)/2 = 0.10$ mm per side

That is correct in intent but *very* tight for a printed part. On resin the bore will come out slightly undersize and will need reaming/filing; on FDM it will be far undersize. Plan on hand-fitting. Do **not** loosen it blindly - a sloppy double-D on a 5.625deg indexing cam translates directly into dot-position error.

4b. Engagement depth is 6 mm, not the 8 mm the comment claims

[SCAD] The bore is subtracted over local z = ?5 .. +4 (`translate([0, 0, -hub_h-1])`), h = `shaft_bore_depth` + 1 = 9). Material exists only over z = ?4 .. +2 (hub 4 mm + disc floor 2 mm). So:

- **actual maximum engagement = 6.0 mm**, and
- the bore is a **through-hole** - it opens onto the top of the disc floor. There is no ceiling and

therefore **no axial stop** locating the cam on the shaft.

The header comment "Hub 4mm below disc + bore 4mm into disc floor = 8mm total engagement" is stale: the disc floor is 2 mm, not 4 mm.

6 mm of double-D on a ~10 mm shaft is *acceptable* engagement for this torque, but the missing axial stop means cam height depends entirely on how far it is pushed on during assembly - with `pin_lift` = 0.8 mm and dot flushness already fought over in v7.2, that is a real risk. Add a 0.5-1.0 mm shoulder at the top of the bore, or a printed

depth gauge.

4c. The vertical stack does not close - SHOWSTOPPER (conditional on motor measurement)

Working in outer-box world z [SCAD]:

feature	z
motor top / bracket face = base-plate underside	41
base plate top	46
cam pocket floor (base_thickness ? cam_pocket_depth)	43
cam disc floor bottom -> top (cam_flat_z)	43 -> 45
cam hub bottom (4 mm below the disc)	39
shaft top, if the shaft is 10 mm [SPEC]	51

Two consequences:

- **The hub is 2 mm too long.** Only 2 mm of space exists between the cam pocket floor (43) and the motor face (41), but hub_h = 4. The hub bottoms out on the motor and the disc sits **2 mm proud of its pocket**, pushing cam_flat_z to 47. Since link_total_h is derived from cam_flat_z = 45, **every one of the six dots ends up ~2 mm permanently proud** - exactly the class of bug v7.2 just fixed for a 0.8 mm offset.

- **The shaft sticks out the top.** With the bore open, a 10 mm shaft reaches z=51, i.e. **6 mm above the cam surface (**4 mm above it even in the mis-seated position**). The dot cluster sits at **x = +/-2.4, y = 0/+/-2.6 - directly over a dia 5 shaft (r=2.5)**. The linkage risers occupy **3.5 to 12.2 mm above the cam surface at exactly those coordinates**. The shaft will foul the dot-2 and dot-5 risers.**

Fixes (all need measurement 6 above first): shorten hub_h to the real gap; counterbore the base plate around the shaft so the hub can drop; trim the motor shaft; or lower the motor by increasing the z-budget. Do not guess - measure the shaft, then re-derive the whole stack in one pass.

5. ULN2003 IN THE 36 x 46 x 16 POCKET - confirmed

[SCAD] outer_box.scad: electronics pocket is cube([36, 46, 16]) centred at x=0,y=0, floor at z=4, top at z=20 (X=36, Y=46). mid_plate.scad adds a through relief slot translate([3, -23, -1]) cube([17, 8, 4]) - 17 mm (x=3..20) by 8 mm (y=-23..-15) - giving **2 mm extra headroom** at that corner only.

configuration	height [ASSUMED, established earlier]	available	verdict
vertical Dupont jumpers plugged into the ULN2003 header	~20 mm	16 mm (18 mm at the relief slot)	FAILS by ~2-4 mm
wires soldered flat to the board	~12 mm	16 mm	FITS, 4 mm spare

Confirmed: solder the wires flat. No vertical jumpers, no exceptions. This already matches docs/WIRING_AND_ASSEMBLY.md.

Three secondary cautions:

- 1 **Plan fit is tight in X.** A ULN2003 driver board is commonly ~35 x 32 mm [ASSUMED - measure].

If its 35 mm side runs along X it has **0.5 mm per side** in a 36 mm pocket. Orient the 35 mm side along **Y** (46 mm) instead.

- 1 **Muscle-board bosses are in the way.** Four dia 4 x 4 mm tall bosses stand on the pocket floor at (+/-14, +/-19). If the ULN2003 is ~35 mm in Y (+/-17.5) it clips the y=+/-19 bosses. The pocket is currently spec'd for ***two different boards*** (the future 34x44 muscle PCB and the off-the-shelf ULN2003) and only one of them can be there at a time - decide which.

- 1 **Floor wire gutters** are cut 2.5 mm deep at x ? [-18,-11] and [11,18] across the whole pocket.

A board laid on the floor will bridge them; fine, but it will not sit flat if it has through-hole tails on the underside.

The mid-plate relief slot is **correctly placed** for a JST/header edge pointed at the (+X, ?Y) corner - it clears the motor collar (whose max x at y=?15 is +0.5) and the boss hole at (26, ?21). Orient the module accordingly.

6. ESP32 DEVKIT IN THE POD - FAILS on length and on USB alignment

[SCAD] `esp32_pod_params.scad / esp32_pod_shell.scad: pod_int_length = 64 ? 2-4 = 56 -> cavity spans x ? [28, +28]. devkit_length = 51.5, devkit_x_offset = +4.`

6a. Length - 1.75 mm interference

Board spans $x = 4 ? 25.75 = 21.75$ to $4 + 25.75 = 29.75$. Cavity ends at **+28**. The board overruns the +X (dock) wall by **1.75 mm**. It will not go in.

The header channels themselves are fine (they span $x = 16 .. +24$, retention walls to ± 25.5 , all inside the cavity) - it is the *board overhanging the sockets* that hits the wall. The antenna grille recess in that wall only starts at $z=23$, far above the board at $z \sim 10.5$, so it does not help.

Fix: reduce `devkit_x_offset` to $\sim +2$ (giving ~ 0.25 mm at each end - still too tight) or lengthen the pod. Realistically `pod_length` needs to go from 64 to ~ 68 , which conveniently also matches the cell footprint. **Measure the board first.**

6b. Width and header pitch - M21 is clone-specific

[SCAD] `devkit_width = 28.0, hdr_row_pitch = 25.4`. The owned board is now confirmed as 15 pins per row / 30 total, USB-C.

[SPEC, still verify owned board] Pin count does not identify the transverse footprint. Commercial 30-pin carriers explicitly support 0.9, 1.0, and 1.1 inch row spacing - 22.86, 25.4, and 27.94 mm. Matching USB-C CH340C listings are approximately 51.5 x 28.5 mm overall.

Therefore the current 25.4 mm channel spacing is plausible, but not proven. Physically measure M21 before printing; a wrong value prevents the DevKit from plugging in at all.

6c. USB cutout - ~ 2 mm too high

[SCAD] `usb_z = pod_floor + hdr_strip_h + 1 = 3 + 8.5 + 1 = 12.5`, cutout 10 wide x 7 tall -> spans $z = 12.5 .. 19.5$.

But `hdr_channel_depth = 1.0` recesses the strips 1 mm into the floor, so the strip top - and therefore the **board underside** - sits at $3 + 8.5 ? 1 = 10.5$, not 11.5. A micro-USB socket sits on the board and stands ~ 2.5 mm tall -> it occupies $z \sim 10.5 .. 13.0$.

Overlap with the cutout: $z = 12.5 .. 13.0$, i.e. **0.5 mm**. The port is essentially behind solid wall. `usb_z` should be ~ 10.0 .

6d. USB cutout size and plug reach

- [SCAD] cutout is 10 x 7 mm in a 4 mm wall.
- [SPEC-ish] a micro-USB plug's *metal shell* is $\sim 7 \times 2.5$ mm, but the **overmoulded body** on a typical cable is $\sim 11\text{-}13$ mm wide and $\sim 7\text{-}9$ mm tall. It will very likely not pass a 10 x 7 hole.
- The board's ?X edge sits at $x = 21.75$ while the wall inner face is at 28, so the plug must reach **6.25 mm inboard** past the wall to seat. Combined with the overmould, this is unlikely to work. Enlarge to $\sim 13 \times 9$ and reduce the standoff distance.

Current source (v7.7): resolved provisionally. The board was moved toward the USB wall, reducing plug reach to 2.25 mm, and the service opening is now **14 x 9 mm**. This deliberately preserves the 9 mm overmould height clearance while adding 1 mm of width. Measure M23 only if the intended cable body exceeds that envelope.

6e. Headroom under the lid - fine

Cavity top at $z = 54$ (`shell_h = 58 ? 4`). Board top ~ 10.5 , tallest DevKit component maybe 5 mm -> $z \sim 15.5$. **~ 38 mm of unused headroom.** No clash with the jack cradle either: the cradle bottom sits at $z = 54 ? 11 ? 1.5 = 41.5$, and it is at $y=+18$ while the board spans $y = \pm 14$.

The pod is enormously over-tall for its contents. Not a fit failure, but worth knowing before printing 58 mm of PETG.

7. BARREL JACK - UNVERIFIED, and the placeholder cradle already collides

[SCAD] `esp32_pod_lid.scad:`

```
jack_body_w = 9.5;    // PLACEHOLDER
jack_body_l = 12;     // PLACEHOLDER
jack_body_h = 11;     // PLACEHOLDER
cradle_t     = 1.5;
```

barrel_jack_dia = 11.5; at (x=-22, y=+18)

The author has flagged these as placeholders. Two things to know:

7a. Even the placeholder cradle does not fit the pod. Cradle outer extent in X = barrel_jack_x ? jack_body_l/2 ? cradle_t = ?22 ? 6 ? 1.5 = **?29.5**. The pod cavity ends at **x = ?28. 1.5 mm of interference - the lid physically cannot close**. Since a real screw-terminal jack is almost certainly *bigger* than the 12 mm placeholder, this gets worse, not better. Either move barrel_jack_x inboard (to about ?20) or shrink the cradle.

7b. The modelled jack is the wrong type. dia 11.5 is sized for a **threaded panel-mount 5.5/2.1 jack with a nut** [SPEC]. Mridul's part is a **yellow screw-terminal barrel socket** - typically a moulded block with two screw terminals and no panel thread, so a round hole gives it nothing to clamp to. That is exactly why the cradle exists, and exactly why the cradle must be built to the real part.

Also still open from the 2026-06-03 breadboard review: **the jack's polarity has never been proven**. Check it with a multimeter before applying adapter power.

Measure before touching this file: see checklist items 14-18 in §10.

8. MAGNETS - the only genuinely measured component. Re-confirm the new batch.

[SCAD] outer_box.scad and esp32_pod_params.scad both use mag_dia = 8.4, mag_depth = 1.2, teardrop profile, 2 per face at y = +/-14, z = 29, in a 4 mm wall (2.8 mm of material left behind each pocket).

[MEASURED, v6.1] docs/SOURCING.md records "NeFeB disc magnets **8mm dia x 1mm thick** - plenty" and the v6.1 CAD notes say they were measured off the real parts. So 8.4 x 1.2 gives 0.2 mm radial and 0.2 mm axial glue clearance. **FITS**.

Caveats:

- The bill line "10x magnets + bearing" does **not** identify a size. Confirm the batch in hand is the same 8 x 1 discs that were measured in v6.1, not a different order.
- Quantity: one cell needs **4** (2 per face x +/-X faces); one pod needs **2**. So a cell+pod pair needs 6, and any second cell needs another 4. 10 covers cell + pod + spares, but not a second cell.
- The pockets open onto the **outer** face, so each magnet sits flush with the outside with 2.8 mm of wall behind it. That is the right way round for holding force. A 1 mm-thick 8 mm disc is still a weak magnet - two per face is not much dock retention, so expect the chain to be light-duty.
- The "bearing" on the bill is **no longer needed** - v7.1 replaced the bearing-ball braille dot with a printed 1.5 mm dome (docs/SOURCING.md §3).
- **Separate gap:** the **cam homing magnet (3 x 2 mm)** is a different part and is **not in hand**. See §2c - and note its pocket is currently broken.

9. DOT INSERT INTO TOP PLATE - FITS (verified numerically)

[SCAD] From mech_layout.scad / dot_insert.scad / top_plate.scad:

feature	insert	plate pocket	clearance
body (square)	insert_body = 11.0	insert_body + insert_fit = 11.2	0.1 mm/side
flange (square)	insert_flange = 15.0	insert_flange + insert_fit = 15.2	0.1 mm/side
body height	insert_h ? insert_flange_h = 2.0	opening runs full 4 mm thickness	ok
flange height	insert_flange_h = 1.2	rebate from z=2.0 to 3.2 = 1.2	exact
total height	insert_h = 3.2	plate_thickness ? finger_pad_depth = 4.0 ? 0.8 = 3.2	flush

- **Glue shelf:** rebate floor at z = 2.0, ring width (15.2 ? 11.2)/2 = **2.0 mm** all round,

area = $15.22 \times 11.22 = 105.6 \text{ mm}^2$. Matches the documented $\sim 106 \text{ mm}^2$.

- **Spring recess ceiling:** spring_bores cut $z = 20.01 \dots 2.0$, i.e. exactly the body height, so the ceiling is the flange underside at insert- $z = 2.0 = \text{linkage-}y = 11.0$. This matches the mech_layout .scad comment "dot DOWN 8.0 -> 11.0 = 3.0 mm". **Consistent.**

- **Dot clearance:** plate_hole_dia 1.7 vs dot_dome_dia 1.5 -> 0.1 mm/side. Correct.

- **Feature envelope:** dots span $x = \pm 2.4$, $y = \pm 2.6$; plus the dia 2.2 spring bore radius that is

$x = \pm 3.5$, $y = \pm 3.7$ - comfortably inside the 11 mm body.

Verdict: FITS. The only real risk is manufacturing, not geometry:

1 The plate is PETG on a 0.4 mm nozzle. A square 11.2 mm hole prints undersize and picks up elephant's foot on the first layers. Expect to **file the opening** or add a 0.15-0.2 mm chamfer.

1 0.1 mm/side is a *press* fit, not a *glue* fit. Thin CA will wick in; epoxy will not - it needs $\sim 0.15\text{-}0.2 \text{ mm}$. Use thin CA on the shelf, per the assembly note in dot_insert .scad.

1 Wipe squeeze-out off the top face before it cures or the domes jam. Already documented.

10. MEASUREMENTS REQUIRED BEFORE ASSEMBLY

Every one of these must be taken with calipers on **the actual part in hand**, not from a datasheet or a product listing. The right-hand column is the CAD symbol it feeds.

Motor - 28BYJ-48 (blocks base_plate, mid_plate, braille_cam, outer_box)

#	Measure	Feeds
1	Can outer diameter	motor_body_diameter (base_plate); collar ID 29.5 (mid_plate)
2	Can height, bottom face to bracket face	base_plate_z = 41 (outer_box)
3	Shaft centre -> can centre distance	motor_body_x_offset = ?8; motor_x_offset = ?8
4	Shaft diameter	cam bore cylinder dia 5.2 ($r=2.6$, braille_cam)
5	Shaft flat-to-flat width	cam bore clip cube([10, 3.2, ...])
6	Shaft length above the bracket face	hub_h = 4, shaft_bore_depth = 8, and the whole z-stack
7	Shaft boss (raised collar) diameter and height	cam hub OD dia 9 ($r=4.5$)
8	Mount-hole centre-to-centre	motor_mount_spacing = 35
9	Mount-hole diameter	motor_mount_hole = 4.3
10	Bracket ear outer width / thickness	base_length = 56 (the 0.35 mm edge wall problem)

Hall sensor (blocks base_plate)

#	Measure	Feeds
11	MH-Sensor module PCB length x width x thickness	hall_pocket_w = 5.3, hall_pocket_d = 4.3 - or the decision to desolder
12	Header height above the PCB (or below, if it must lie flat)	hall_pocket_h = 3
13	If desoldering: bare TO-92 body width x depth x height, and lead pitch	hall_pocket_w/d/h, hall_wire_w/d

Barrel jack (blocks esp32_pod_lid)

#	Measure	Feeds
14	Jack body X length (along the pod's X axis)	jack_body_l = 12 (PLACEHOLDER)
15	Jack body Y width	jack_body_w = 9.5 (PLACEHOLDER)

#	Measure	Feeds
16	Jack body Z height below the lid, including screw terminals	jack_body_h = 11 (PLACEHOLDER)
17	Barrel outer diameter at the face (the part that goes through the lid)	barrel_jack_dia = 11.5
18	Does it have a panel thread + nut? Thread dia and nut AF if so	decides hole vs cradle strategy
19	Polarity (multimeter: which terminal is centre-pin positive)	wiring, not CAD - but do it before power-on

ESP32 DevKit (blocks esp32_pod_shell, esp32_pod_params)

#	Measure	Feeds
20	Board length	devkit_length = 51.5
21	Board width	devkit_width = 28.0
22	Pin-row centre-to-centre pitch (critical - 30-pin vs 38-pin)	hdr_row_pitch = 25.4
23	Pin count per row	hdr_strip_len = 40
24	USB socket: width, height, and height of its underside above the board	usb_w = 10, usb_h = 7, usb_z = 12.5
25	Your USB cable's overmould width x height	usb_w, usb_h
26	Female header strip: body width and height with a board seated	hdr_strip_w = 2.7, hdr_strip_h = 8.5

ULN2003 driver module (blocks outer_box, mid_plate)

#	Measure	Feeds
27	Board length x width	elec_pocket 36 x 46 (outer_box)
28	Total height with wires soldered flat	elec_pocket_h = 16
29	Height and position of the tallest connector	mid_plate relief slot [3, ?23] cube([17, 8])
30	Mounting-hole positions, if you intend to bolt it	mb_boss_positions = (+/-14, +/-19)

Magnets and switches

#	Measure	Feeds
31	Docking magnet diameter x thickness (re-confirm the new batch)	mag_dia = 8.4, mag_depth = 1.2
32	Homing magnet diameter x thickness (must be bought - 3x2 mm)	magnet_dia = 3.0, magnet_depth = 2.0
33	Tactile switch body W x D, total height including plunger, plunger travel	sw_body = 6.4, sw_depth = 5.5, cage depth 7.2, nav_shaft_len = 4.5
34	Switch lead spread and lead orientation	switch cage fin spacing

Pogo pins (not audited here - no dimensions exist in the CAD at all)

#	Measure	Feeds
35	Pogo carrier board width x height x thickness	pogo_carrier_w = 12, pogo_carrier_h = 10, pogo_carrier_d = 2 - all explicitly marked "measure real part!"
36	Pogo pin free length and compressed length	pogo_slot_cube([6, 10, 8]), pogo_pad_recess = 1

11. BOTTOM LINE

Do not order resin yet, and do not attempt a full assembly.

Blocking failures that are certain from the source, independent of any measurement:

- 1 The hall sensor pocket **does not exist** - it is entirely inside the cam pocket at the same depth (§2a).
- 2 The cam homing-magnet pocket is a **through-hole that craters cam tracks 2 and 3** (§2c).
- 3 The motor's **right mounting screw hole opens into the spring cavity** - one-screw motor (§3a).
- 4 The motor's **left screw hole has a 0.35 mm wall** to the plate edge (§3b).
- 5 The **ESP32 board overruns the pod cavity by 1.75 mm** (§6a).
- 6 The **USB cutout is ~2 mm above the port** (§6c).
- 7 The **jack cradle overruns the pod cavity by 1.5 mm**, with placeholder dimensions (§7a).
- 8 The **cam z-stack does not close**: hub 2 mm too long, shaft ~6 mm too long for the open bore,

putting all six dots permanently proud and the shaft into the dot-cluster risers (§4c).

What actually passes: the **dot insert / top plate** interface (§9), the **docking magnets** (§8), and the **ULN2003 in the pocket provided its wires are soldered flat** (§5).

The single highest-value next action remains the one in the handoff: **take the 36 measurements above**, starting with the motor, then re-derive the z-stack in one pass. Nothing downstream is trustworthy until the motor is a measured object rather than a datasheet.